

Design & Analysis of Algorithms

Lab 1: Algorithm Performance Analysis Workbench

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Q1. 1.4 Algorithm Execution Observation Table (5 marks)

import math

```
def generate_execution_observation_table(sizes):
    # Helper function to compute recursive Fibonacci function calls
    def recursive_fib_calls(n):
        if n <= 1:
            return 1
        a, b = 1, 1
        for _ in range(2, n + 1):
            a, b = b, a + b + 1
        return b

    # Header initialization
    output = [
        "Algorithm Execution Observation Table",
        "InputSize RecursiveFactorial IterativeFactorial RecursiveFibonacci"
        " IterativeFibonacci LinearSearch BinarySearch BubbleSort"
        " InsertionSort",
    ]

    # Calculate metrics for each input size n
    for n in sizes:
        rec_fact = n + 1
        iter_fact = n
        rec_fib = recursive_fib_calls(n)
        iter_fib = n
        lin_search = n
        bin_search = math.floor(math.log2(n)) + 1
        bubble_sort = n * (n - 1) // 2
        insertion_sort = n * (n - 1) // 2

        row = (
            f"{n} {rec_fact} {iter_fact} {rec_fib} {iter_fib} {lin_search}"
            f" {bin_search} {bubble_sort} {insertion_sort}"
        )
        output.append(row)

    return output
```